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| **Fiscal Year:** | FY 2017 | **Task Last Updated:** | FY 03/01/2017 |
| **PI Name:** | Alwood, Joshua Ph.D. | | |
| **Project Title:** | 2012 Presidential Early Career Award for Scientist and Engineers (PECASE) | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Musculoskeletal Biology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
| **PI Email:** | joshua.s.alwood@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | NASA CENTER | **Phone:** | 650-604-1490 |
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| **Zip Code:** | 94035 | **Congressional District:** | 18 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Space Biology PECASE AWARD |
| **Start Date:** | 04/01/2015 | **End Date:** | 09/30/2018 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | 3 |
| **No. of Bachelor's Candidates:** | 5 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Sato, Kevin | **Contact Phone:** | 650-604-1104 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | Internal Project | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | This Presidential Early Career Award for Scientist and Engineers (PECASE) award consist of two mini-projects.<br>Project 1. Epigenetic Regulation During Simulated Spaceflight: MicroRNA Expression, Histone Deacetylase Status, and Methylation-Specific Sequencing<br><br>Description and Aim: This work complements the methylation work performed in the current Space Biology New Investigator Grant ("Bone Loss During Simulated Weightlessness: The Role of Osteoclasts," ends April 2015), by allowing ‘omics approaches to cover larger swaths of the genome (e.g., whole genome sequencing and RNA sequencing). Additionally, this project will extend the type of epigenetic mechanisms to be studied beyond DNA methylation. Specifically, we will analyze microRNA expression and status of histone acetylation. Hypotheses: Removal | | |

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| **Task Description:** | of mechanical loading or, oppositely, elevations of mechanical loading induce pro-bone resorbing or pro-bone forming changes in bone cell gene expression, respectively, via epigenetic mechanisms.<br><br>Project 2. Skeletal Sex Hormone Receptors and Simulated Spaceflight<br><br>Description and Aim: This work complements an upcoming space mission (Principal Investigator (PI)) : Joe Tash) and will define the skeletal response of male or female mice to simulated long-duration spaceflight in the context of sex hormones and their receptor levels in each compartment of the femur (cancellous and cortical). The assays used will include RNA-Seq, protein quantification in bone and blood, and immunostaining of histological sections of the femur for sex hormone and receptor levels. Hypotheses: Mechanical disuse reduces the levels of circulating sex hormones and their receptor levels in bone. These changes contribute to skeletal deterioration during long-duration spaceflight. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This project addresses multiple Research Recommendations for Animal Experiments and Gaps in Knowledge stated in the Space Biology Science Plan and the National Academy of Sciences Decadal Survey, Recapturing Space: Life and Physical Sciences for a New Era of Exploration. The knowledge generated by this research queries the molecular responses to the high-turnover bone loss that astronauts experience with simultaneous weightlessness and exercising. This unique environment has similarities to Earth-bound sedentary or bed-ridden individuals. Additionally, this work investigates fundamental molecular mechanisms, specifically, non-coding RNA and DNA methylation responses to simulated spaceflight, that may improve our fundamental understanding of skeletal regulation and the organism's response to spaceflight. |
| **Task Progress:** | In the second year of the award period, progress has been made on each project listed in the Task Description.<br>For Project 1, one experiment and one new assay have been conducted. The experiment investigated the effects of musculoskeletal disuse, as a model of microgravity exposure, on non-coding microRNAs within the skeleton. Assessment has taken place with PCR-based array. As a new assay, we successfully sequenced the DNA methylation levels within bone, as a small pilot study.<br><br>For Project 2, we investigated the effects of musculoskeletal disuse on circulating sex-hormone levels in the blood serum. Additionally, we have been awarded vertebral tissue from the Rodent Research 1 mission. Microcomputed tomography scans of the lumbar, sacral, and caudal vertebrae have been completed.<br><br>Finally, two brand new collaborations have commenced, which investigate different hypotheses. Specifically, one project aims to investigate the effects of ionizing-radiation exposure on bone's material properties and the other aims to investigate the role of cytoskeletal elements in the skeletal response to simulated weightlessness. |
| **Bibliography Type:** | Description: (Last Updated: 05/25/2026) |
| **Abstracts for Journals and Proceedings** | Dubée PB, Scott RT, Thomas NH, Pendleton MN, Alwood JS. "Quantifying Cancellous Bone Structural Changes from Microgravity: Axial Skeleton Results from the Rodent Research-1 Mission." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016.<br>32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. https://asgsr.org/ , Oct-2016 |
| **Abstracts for Journals and Proceedings** | Peter J, Shirazi-Fard Y, Alwood J, Schreurs A-S, Terada M, Tahimic C, Globus R. "Short-Term Effects of Simulated Space Radiation and Microgravity on Cancellous Bone." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016.<br>32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. https://asgsr.org/ , Oct-2016 |
| **Abstracts for Journals and Proceedings** | Thomas NJ, Choi CY, Alwood JS. "Are micro-RNAs the missing link between sexual reproductive health and bone formation in space?" 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016.<br>32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. https://asgsr.org , Oct-2016 |
| **Abstracts for Journals and Proceedings** | Thomas NJ, Choi CY, Alwood JS. "Skeletal micro-RNA Responses to Simulated Weightlessness." International Space Station (ISS) Research and Development Conference 2016, San Diego, California, July 12-16, 2016.<br>International Space Station (ISS) Research and Development Conference 2016, San Diego, California, July 12-16, 2016. https://ntrs.nasa.gov/search.jsp?R=20160009015 , Jul-2016 |
| **Abstracts for Journals and Proceedings** | Pendleton MM, Alwood JS, Sibonga JD, O'Connell GD, Keaveny TM. "Effects of Ionizing Radiation on Ex-Vivo Fatigue Mechanics of Mouse Vertebra." ASBMR 2016 (American Society for Bone and Mineral Research), Atlanta, Georgia, September 16-19, 2016.<br>Journal of Bone and Mineral Research. 2016;31 (Suppl 1). , Sep-2016 |
| **Abstracts for Journals and Proceedings** | Scott RT, Nalavadi MO, Anand S, Shirazi-fard Y, Alwood JS. "Zoledronate prevents simulated weightlessness-induced bone loss in the cancellous compartment while blunting the efficacy of a mechanical loading countermeasure." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017.<br>2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017 |